

GED Math Assignment 6 Solutions.

1. Total number of yards =

$$\begin{aligned} & \text{number of yards per sofa} \times \text{number of sofas} \\ &= 12 \text{ yards/sofa} \times 3 \text{ sofas} \\ &= 36 \text{ yards.} \end{aligned}$$

2. # gallons needed = $\frac{3500 \text{ ft}^2}{350 \frac{\text{ft}^2}{\text{gal.}}}$

$$= 10 \text{ gallons.}$$

3. # miles farther = Kate's distance - Sarah's distance

$$\begin{aligned} &= 30 - 25 \\ &= 5 \text{ miles} \end{aligned}$$

4. $d = rt$

$$\begin{aligned} &= 65 \frac{\text{miles}}{\text{hr}} \times 4 \text{ hr} \\ &= 260 \text{ miles} \end{aligned}$$

5. (over) * Some research needed for this question - a loonie is worth \$1 and a toonie is worth \$2.

#5. Let x be the number of loonies and y be the number of twonies.

$$\textcircled{1} \quad x + y = 23$$

$$- \textcircled{2} \quad (x + 2y = 36)$$

$$-y = -13$$

$$y = 13 \rightarrow \text{into } \textcircled{1}$$

$$x + 13 = 23$$

$$x = 23 - 13$$

$$= 10$$

$\therefore$ There are
10 loonies and
13 twonies.

6. Diane's rate = $\frac{1}{8}$ jobs/h
Brenda's rate = $\frac{1}{7}$ jobs/h

so if x = time to complete the job,

$$\frac{1}{8}x + \frac{1}{7}x = 1 \quad (\text{multiply by } 7 \times 8)$$

$$7x + 8x = 56$$

$$15x = 56$$

$$x = \frac{56}{15}$$

$$= 3.73$$

$\therefore$ The job will take

3.73 hours to
complete, or

3 hours 44 minutes.